

The 2.5G technologies brought multimedia to our mobile phones, but the data transfer rates have still left a lot to be desired.

The Road To 3G—The Third Generation Partnership Projects (3GPP and 3GPP2)

The Third Generation of mobile telephony is supposed to usher in a uniform standard for cellular networks worldwide, resulting in true mobility—your handset will connect you to anyone, anywhere, wherever you are. Third Generation Technology is based on CDMA, but due to the insane popularity and sheer coverage of GSM networks, things aren't really moving as planned.

The Third Generation Partnership Project (3GPP) is a collaboration of telecom associations from different parts of the world to make a globally applicable third generation system based on GSM networks. These same associations have also collaborated under the Third Generation Partnership Project—2 (3GPP2), which sets standards for a third generation system based on CDMA.

The associations are:

- Association of Radio Industries and Businesses—ARIB, China
- Telecom Technology Community—TTC, Japan
- China Communication Standards Association—CCSA, China
- Telecom Industries Association—TIA, North America
- Telecom Technologies Association—TTA, Korea

Before a cellular network can be called Third Generation, it has to meet the norms set by the International Telecom Union (ITU)'s IMT 2000 Specification, proposed by the 3GPP. Some of the requirements are:

- Data transfer rates of 144 kbps for highly mobile traffic (moving cars, etc.), 384 kbps for pedestrian traffic and 2 Mbps for indoor traffic.
- A common billing system, where usage information and user profiles are shared between operators
- The ability to deliver fixed and variable bit-rate multimedia to the mobile phone